외계행성 대기
Exoplanetary Atmospheres

Lecture 8: Vertical Thermal Structure of a Planetary Atmosphere II
2026 November 2 (Monday)

Based on: Chapter 9 of Seager, *Exoplanet Atmospheres* (2010), and Chapter 4 of Heng, *Exoplanetary Atmospheres: Theoretical Concepts and Foundations* (2017)

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Outline – Lecture 8

Surface Temperature in Simple Greenhouse Models (Continued)

Convection versus Radiation

The Radiative-Equilibrium Temperature Profile

The Adiabatic Temperature Profile

The One-Dimensional T–P Profile

Temperature Retrieval

Summary

A leaky greenhouse: the setup

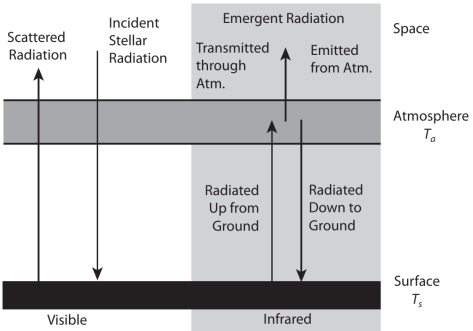


Figure 1: The leaky greenhouse: the layer absorbs only a fraction α of the infrared. Seager (2010), Fig. 9.5.

The simple greenhouse atmosphere traps *all* infrared radiation from the ground. A slightly more realistic "leaky" layer is partly transparent (optically thin) in the infrared:

- the atmosphere layer absorbs a fraction $\alpha \in (0,1)$ of the upwelling surface emission and transmits $1-\alpha$ to space;
- by **Kirchhoff's law** its emissivity equals its absorptivity$\times$B, so it emits the flux $\mathscr{F}_{\text{emiss}}\uparrow = \alpha\sigma T_a^4$ upward and $\mathscr{F}_{\text{emiss}}\downarrow = \alpha\sigma T_a^4$ downward;
- starlight is again assumed to pass the layer and heat the ground. Again, the starlight is assumed not to be scattered back to space and all of the radiation reaches the planet surface.

A leaky greenhouse: the solution

Two contributions now leave the planet (Fig. 9.5):

- emission from the atmosphere layer;
- surface emission that **leaks** through the layer.

Two energy-balance equations (top and surface) determine the two unknowns T_s and T_a:

$$\begin{aligned} \text{top of the atmosphere:} \quad & \mathscr{F}_* \downarrow \quad = \mathscr{F}_{\text{atm}} \uparrow + (1-\alpha)\,\sigma T_s^4 \uparrow, \\ \text{surface:} \quad & \mathscr{F}_* \downarrow + \mathscr{F}_{\text{atm}} \downarrow = \sigma T_s^4 \uparrow . \end{aligned}$$

Here, $\mathscr{F}_* \downarrow = \sigma T_e^4$ is the stellar flux absorbed by the ground and $\mathscr{F}_{\text{atm}} \uparrow = \mathscr{F}_{\text{atm}} \downarrow = \sigma T_a^4$ is the flux emitted by the atmosphere layer.
Eliminating $\mathscr{F}_{\text{atm}} \downarrow = \mathscr{F}_{\text{atm}} \uparrow$: $2T_e^4 = (2-\alpha)\,T_s^4$. Then $T_a^4 = T_s^4 - T_e^4 = \alpha T_e^4/(2-\alpha)$:

$$\boxed{T_s = \left(\frac{2}{2-\alpha}\right)^{1/4} T_e}, \qquad \boxed{T_a = \left(\frac{\alpha}{2-\alpha}\right)^{1/4} T_e = \left(\frac{\alpha}{2}\right)^{1/4} T_s}.$$

Check: the atmosphere layer absorbs $\alpha\sigma T_s^4$ and emits $2\sigma T_a^4$, so $T_s^4 = (2/\alpha)T_a^4$.

A leaky greenhouse: what it means

$$T_s = \left(\frac{2}{2-\alpha}\right)^{1/4} T_e, \qquad T_a = \left(\frac{\alpha}{2-\alpha}\right)^{1/4} T_e$$

- **A partly transparent layer reduces the greenhouse warming.**
- $\alpha = 1$: the simple greenhouse, $T_s = 2^{1/4} T_e$ (a special case of the leaky model).
- $\alpha \to 0$: transparent layer or no atmosphere, $T_s = T_e$ and $T_a = 0$ (no atmosphere).
- Since $1 \le 2-\alpha \le 2$: always $\boxed{T_s \ge T_e \ge T_a}$. The ground is warmer than the emission temperature, and the atmospheric layer is colder than it.
- Earth: $T_s = 288\,\mathrm{K}$ with $T_e = 255\,\mathrm{K}$ needs $2/(2-\alpha) = (288/255)^4 \approx 1.63$, i.e. $\alpha \approx 0.77$.
- **A single layer can never give more than $T_s = 2^{1/4} T_e$. Venus needs many opaque layers** (next slide). Extended to many layers, with the right α for each layer at every wavelength, and with molecules and clouds, the model approaches a multilayer **radiative-equilibrium** model.

Many ‘non-leaky’ layers: towards radiative equilibrium

Stack n layers above the surface (Seager’s exercise 3). Assumptions:

- All layers are **identical in their optical properties**: transparent in the visible (starlight heats only the ground) and **fully opaque in the infrared** ($\alpha = 1$), each emitting as a black body up and down.
- Only radiation carries energy (no scattering, convection or conduction). Thickness, density and pressure never enter; the layers differ *only* in temperature, which is an output of energy balance, not an input.

Number the layers $k = 1, \ldots, n$ from the top, with the surface as $k = n+1$ and $T_0 = 0$ (no infrared from space). Layer k absorbs the emission of its two neighbors and emits from both faces, $2\sigma T_k^4 = \sigma T_{k-1}^4 + \sigma T_{k+1}^4$, with $\sigma T_1^4 = \sigma T_e^4$ (the top layer is what space sees). T_k^4 is therefore linear in k:

$$T_k^4 = k\,T_e^4 \quad \Longrightarrow \quad \boxed{T_s = (n+1)^{1/4}\,T_e} \quad \text{(for } n \text{ opaque layers with } \tau_k \approx 1 \text{ and } \tau_{\text{tot}} \approx n\text{)}.$$

- Venus: $(T_s/T_e)^4 = (730/230)^4 \approx 100$, i.e. the equivalent of ~ 100 opaque layers — an infrared optical depth of order 100 above the surface. The massive CO_2 atmosphere (92 bar) is that opaque.
- With $n \to$ many layers of appropriate absorptivity, at all wavelengths, with molecules and clouds, these models become a **multilayer radiative-equilibrium model**: the subject of Lecture 8.
- Each “layer” is one unit of infrared optical depth, not a fixed geometric thickness. The model is a discrete gray atmosphere with $\tau \approx n$, and $T_k^4 \propto k$ mirrors the gray $T^4 \propto \tau$.
- For leaky layers ($\alpha < 1$) the recursion changes and $T_s < (n+1)^{1/4} T_e$; already for $n = 1$, $T_s^4 = 2T_e^4/(2-\alpha)$.
- The one-layer ratio $T_a/T_s = 2^{-1/4}$ reappears in the gray model as the skin temperature $T(0) = 2^{-1/4} T_{\text{eff}}$.

Why T_{eq} is not enough: a myriad of effects

Without data, an exoplanet is labeled by its equilibrium temperature, computable without knowing anything about its atmosphere. The Solar System shows how misleading that can be: Earth's T_{eq} is below freezing, yet it has liquid water; Venus's T_{eq} is barely higher, yet its surface is about 700 K. Real atmospheres contain many opacity sources that warm or cool them (Heng, Ch. 4):

- **Greenhouse warming:** gases such as H_2O, CO_2 and CH_4 absorb the planet's *thermal* emission.
- **Anti-greenhouse cooling:** enhanced absorption, or scattering, of *starlight* high in the atmosphere.
- **Scattering greenhouse:** thermal emission is scattered back toward the planet.
- **Aerosols and clouds** combine all of these, and can cool or heat depending on their properties and where they sit.

The aim, following Pierrehumbert, is to understand this richness with simple, clear models that isolate one effect at a time — before, or instead of, a full simulation.

Convection versus Radiation: Which process carries the energy?

Energy can be transported by **radiation, convection, or conduction**. In the relatively dense lower atmospheres, thermal conduction is negligible. Energy takes the **path of least resistance**: if the opacity is low, then the energy is most likely to be transported by photons; if the opacity is high, radiation is impeded and energy is more likely to be transported by convection.

How the temperature gradient determines whether or not convection occurs:
Stability of a displaced parcel. Lift a parcel slightly: it expands *adiabatically* to the ambient pressure, keeping its own temperature and density. (adiabatic means no heat is exchanged with the surroundings.)

- if it is colder (denser) than its new surroundings: it sinks back. The atmosphere is **stable** against convection.
- if it is warmer (lighter) than its surroundings: buoyancy keeps it rising. The atmosphere is **unstable**; convection occurs.

A situation is **stable** if, after a disturbance, the system returns to its original state. Which case occurs is decided by comparing two temperature gradients (next slide).

The criterion for convection

Two temperature gradients compete:

- the **adiabatic** gradient $(\mathrm{d}T/\mathrm{d}z)_{\mathrm{ad}}$, followed by the rising parcel;
- the **atmospheric** gradient $(\mathrm{d}T/\mathrm{d}z)_{\mathrm{atm}}$ of the surroundings.

Start the parcel at height z with the ambient temperature T_0 and lift it by Δz. Both gradients are negative where T falls with height, so

$$T_{\mathrm{parcel}} = T_0 - \left|\frac{\mathrm{d}T}{\mathrm{d}z}\right|_{\mathrm{ad}} \Delta z, \qquad T_{\mathrm{atm}} = T_0 - \left|\frac{\mathrm{d}T}{\mathrm{d}z}\right|_{\mathrm{atm}} \Delta z.$$

The parcel has the ambient pressure, so by the ideal gas law ($\rho = \mu_m m_{\mathrm{H}} P/kT$, same μ_m) **the warmer gas is lighter**. The parcel keeps rising if $T_{\mathrm{parcel}} > T_{\mathrm{atm}}$, so **the criterion for convection is a criterion for buoyancy**:

$$\boxed{\left|\frac{\mathrm{d}T}{\mathrm{d}z}\right|_{\mathrm{atm}} > \left|\frac{\mathrm{d}T}{\mathrm{d}z}\right|_{\mathrm{ad}}}$$

(Schwarzschild criterion; Seager Eq. 9.23).

- Unstable: the atmosphere cools with height *faster* than the adiabat; the adiabat is the "shallower" gradient.
- Stable: the atmosphere cools more slowly than the adiabat, is isothermal, or warms with height (an inversion). Note that $T_{\mathrm{atm}} = T_0 + |\mathrm{d}T/\mathrm{d}z|_{\mathrm{atm}} \Delta z$ when temperature increases with height.

Stable and unstable profiles

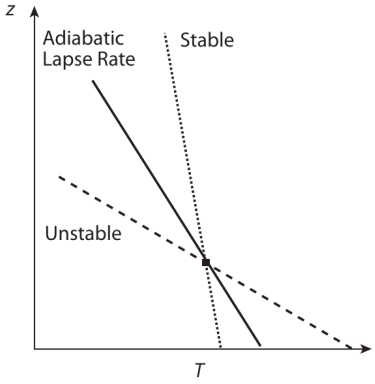


Figure 2: Profiles stable (dotted) or unstable (dashed) against convection, relative to the adiabat. Seager (2010), Fig. 9.6.

In the left figure (Fig. 9.6, T horizontal, z vertical), starting from a common point on the adiabat (solid line):

- a profile that loses *less* temperature per unit height than the adiabat: a lifted parcel ends up colder and denser than its surroundings and sinks back: **stable**;
- a profile that loses *more* temperature per unit height than the adiabat: the lifted parcel is warmer and lighter and keeps rising: **unstable**.

An unstable profile cannot persist: convection mixes the layer and drives its gradient back towards the adiabat. Convective regions therefore follow the adiabat closely (Lecture 8).

To decide *when* a layer is unstable we need the actual sizes of the two gradients (next slide).

When is the adiabatic gradient small, or the radiative gradient large?

Which $(\mathrm{d}T/\mathrm{d}z)_{\mathrm{atm}}$ goes into the criterion? We ask whether a layer in which *radiation alone* carries the flux would be stable. Its gradient is then the radiative-equilibrium one, $(\mathrm{d}T/\mathrm{d}z)_{\mathrm{atm}} \approx (\mathrm{d}T/\mathrm{d}z)_{\mathrm{rad}}$. If that gradient is unstable, convection sets in and the actual gradient relaxes towards the adiabat.

The two gradients to compare (Seager Eqs. 9.24–9.25):

$$\left(\frac{\mathrm{d}T}{\mathrm{d}z}\right)_{\mathrm{ad}} = -\frac{g}{c_p}, \qquad \left(\frac{\mathrm{d}T}{\mathrm{d}z}\right)_{\mathrm{rad}} = -\frac{3}{16}\,\frac{\kappa F_{\mathrm{int}}}{\sigma_{\mathrm{SB}} T^3} \quad \Longrightarrow \quad \text{convection if} \quad \frac{3\kappa F_{\mathrm{int}}}{16\sigma_{\mathrm{SB}} T^3} > \frac{g}{c_p}.$$

The adiabat is derived later. The radiative gradient is the Rosseland diffusion result, with κ the mean extinction per unit length (α_R).

- **Small adiabatic gradient = large c_p.** c_p is the heat capacity, the energy needed to heat unit mass of gas by 1 K at constant pressure. Where hydrogen ionizes, the added heat goes into removing electrons rather than into the kinetic energy of the gas, so c_p becomes large and g/c_p small: ionization zones of stars, and the hot interiors of giant planets like Jupiter, where hydrogen starts to ionize above ~ 4000 K.
- **Large radiative gradient = large F_{int} or large κ.** The diffusion formula holds only deep in the atmosphere, but it shows the trend. Large fluxes drive the convective interiors of giant planets; in the atmospheres the flux alone is usually not large enough, and the **opacity** is the main driver (next slide).

Seager gives the units of c_p as $\mathrm{J\,kg^{-1}\,s^{-1}}$; they are $\mathrm{J\,kg^{-1}\,K^{-1}}$.

Convection in planets

- **Opacity drives convection.** Opacity grows with the density of absorbers, so the deep, optically thick layers are almost always convective, while radiation dominates higher up. Opacity can also change with T and P as the composition changes.
- **Jupiter:** a radiative zone in the upper atmosphere and a convective zone in the lower atmosphere — but perhaps another radiative zone underneath, where the mean opacity has a minimum ($T \approx 1300$–$1800\,\mathrm{K}$; Guillot et al. 1994).
- **Hot Jupiters:** strong irradiation makes the dayside temperature profile nearly isothermal or even inverted, so the radiative zone extends *deep* into the atmosphere.
- **Rocky planets with thin atmospheres:** most visible-wavelength starlight is absorbed at the planet surface, which becomes much hotter than the air above. The large upward flux from the ground makes the radiative gradient steep, and a **convective layer** forms just above the surface — on Earth most of the troposphere.
- **Inversions are very stable:** a lifted parcel cools as it expands while its surroundings get warmer with height, so it sinks back strongly. Stratospheres do not convect. Most of the solar system planets have stratospheres, heated by UV absorption in ozone (Earth) or hydrocarbons (Jupiter, Saturn, Titan).

Radiative equilibrium

If energy is carried only by radiation, no energy is created or destroyed in a layer: what enters a volume must leave it. Equivalently, the net flux is constant,

$$\frac{\mathrm{d}F(\tau)}{\mathrm{d}\tau} = 0.$$

Total absorption per volume is $\int\!\!\int \kappa_\nu I_\nu \,\mathrm{d}\Omega\,\mathrm{d}\nu = 4\pi \int \kappa_\nu J_\nu \,\mathrm{d}\nu$; with isotropic emission and $\varepsilon_\nu = \kappa_\nu S_\nu$, total emission is $4\pi \int \kappa_\nu S_\nu \,\mathrm{d}\nu$. Hence

$$\boxed{\int_0^\infty \kappa_\nu \left(J_\nu - S_\nu\right) \mathrm{d}\nu = 0}$$

- Absorption and emission balance *in total*, not at each frequency: a layer can absorb in the visible and emit in the infrared.
- The constraint involves J_ν, so it must be solved *together* with the transfer equation. The opacity's frequency dependence enters directly, and an inversion can arise naturally from the solution.
- The thermal (infrared) flux at the top is $\sigma T_{\mathrm{int}}^4 + \mu_0 I_0$: the interior heat plus the reradiated starlight. Interior heat comes from slow loss of formation energy (Jupiter emits about 1.7 times what it absorbs) or radioactivity (Earth); for Earth and close-in planets the stellar term dominates.

No general analytic solution exists; the **gray** atmosphere gives one.

Gray atmosphere heated from below

Assume LTE, no scattering, and frequency-independent opacity (gray). The energy comes from below only: from the interior of a giant planet ($T_{\rm int} = T_{\rm eff}$), or from a ground heated by sunlight that the atmosphere does not absorb ($T_{\rm int} = T_s$).

Step 1: radiative equilibrium in the gray case. With the frequency-integrated quantities $J(\tau) = \int_0^\infty J_\nu \,\mathrm{d}\nu$, $I(\tau,\mu) = \int_0^\infty I_\nu \,\mathrm{d}\nu$, $S(\tau) = \int_0^\infty S_\nu \,\mathrm{d}\nu$, and $\kappa_\nu = \kappa$, the condition $\int \kappa_\nu (J_\nu - S_\nu)\,\mathrm{d}\nu = 0$ becomes $J = S$. In LTE, $S = B$ integrated over frequency, so

$$J(\tau) = S(\tau) = B\big(T(\tau)\big) = \frac{\sigma T^4}{\pi}.$$

Step 2: the gray transfer equation. Integrating $\mu\,\mathrm{d}I_\nu/\mathrm{d}\tau = I_\nu - S_\nu$ over frequency and using $S = J$,

$$\mu \frac{\mathrm{d}I(\tau,\mu)}{\mathrm{d}\tau} = I(\tau,\mu) - J(\tau).$$

The goal is to show $T^4 \approx \frac{3}{4} T_{\rm int}^4 (\tau + \frac{2}{3})$. The method (Seager follows Hansen 2008) is to take **moments**: multiply by powers of μ and integrate over solid angle, $\mathrm{d}\Omega = \mathrm{d}\mu\,\mathrm{d}\phi$ with $-1 \le \mu \le 1$, $0 \le \phi < 2\pi$.

Gray atmosphere heated from below: moments

Recall the moments $J = \frac{1}{4\pi}\oint I\,\mathrm{d}\Omega$, $F = \oint I\mu\,\mathrm{d}\Omega$, $K = \frac{1}{4\pi}\oint I\mu^2\,\mathrm{d}\Omega$.

Step 3: zeroth moment ($\times \frac{1}{4\pi}\oint \mathrm{d}\Omega$):

$$\frac{1}{4\pi}\frac{\mathrm{d}F}{\mathrm{d}\tau} = J - J = 0 \quad \Longrightarrow \quad F(\tau) = F = \text{const.}$$

This is the flux constancy of radiative equilibrium. All the flux comes from below, so $F = \sigma T_{\text{int}}^4$ at every level.

Step 4: first moment ($\times \frac{1}{4\pi}\oint \mu\,\mathrm{d}\Omega$; the term $J\oint \mu\,\mathrm{d}\Omega$ vanishes because J is isotropic):

$$\frac{\mathrm{d}K}{\mathrm{d}\tau} = \frac{F}{4\pi} \quad \Longrightarrow \quad K(\tau) = \frac{F}{4\pi}\,\tau + c,$$

with an integration constant c, since F does not depend on τ.

Step 5: close the system with the Eddington approximation $K = \frac{1}{3}J$ (Lecture 7):

$$J(\tau) = \frac{3F}{4\pi}\,\tau + 3c.$$

The moment equations alone cannot fix c; it needs a boundary condition at the top of the atmosphere.

Gray atmosphere heated from below: the boundary condition

Step 6: the emergent flux. At $\tau = 0$ nothing comes in from above, and the outgoing intensity is the source function attenuated along the slant path, $I(0,\mu) = \int_0^\infty S(\tau)\,\mathrm{e}^{-\tau/\mu}\,\mathrm{d}\tau/\mu$. Then $F(0) = 2\pi\int_0^1 I(0,\mu)\,\mu\,\mathrm{d}\mu$ gives

$$F(0) = 2\pi\int_0^1\int_0^\infty S(\tau)\,\mathrm{e}^{-\tau/\mu}\,\mathrm{d}\tau\,\mathrm{d}\mu.$$

Insert $S = J = a\tau + b$ with $a = 3F/4\pi$, $b = 3c$, and use $\int_0^\infty \tau\,\mathrm{e}^{-\tau/\mu}\mathrm{d}\tau = \mu^2$, $\int_0^\infty \mathrm{e}^{-\tau/\mu}\mathrm{d}\tau = \mu$:

$$F(0) = 2\pi\int_0^1 \left(a\mu^2 + b\mu\right)\mathrm{d}\mu = 2\pi\left(\frac{a}{3} + \frac{b}{2}\right) = \frac{F}{2} + \pi b.$$

Flux constancy requires $F(0) = F = \sigma T_{\mathrm{int}}^4$, so $\pi b = F/2$, i.e. $3c = F/2\pi = \sigma T_{\mathrm{int}}^4/2\pi$.

Step 7: the temperature profile. Now $J = \frac{3F}{4\pi}\left(\tau + \frac{2}{3}\right)$. Setting $J = \sigma T^4/\pi$ and $F = \sigma T_{\mathrm{int}}^4$:

$$\frac{\sigma T^4}{\pi} = \frac{3\sigma T_{\mathrm{int}}^4}{4\pi}\left(\tau + \frac{2}{3}\right) \quad\Longrightarrow\quad \boxed{T^4 \approx \frac{3}{4}\,T_{\mathrm{int}}^4\left(\tau + \frac{2}{3}\right)}$$

The "$\approx$" comes from the Eddington closure $K = J/3$; the exact gray solution replaces $2/3$ by a slowly varying function $q(\tau)$, the Hopf function, with $q(0) = 1/\sqrt{3} \approx 0.577$ and $q(\infty) \approx 0.710$.

Gray atmosphere heated from below: what it tells us

$$T^4 \approx \frac{3}{4} T_{\rm int}^4 \left(\tau + \frac{2}{3} \right)$$

- **Top of the atmosphere:** $T(0)^4 = T_{\rm int}^4/2$, the same $2^{-1/4}$ ratio as the simple greenhouse (Lecture 7).
- **Gradient:** differentiating with $\mathrm{d}\tau = -\kappa\,\mathrm{d}z$,

$$\frac{\mathrm{d}T}{\mathrm{d}z} = -\frac{3}{16}\,\kappa\,\frac{T_{\rm int}^4}{T^3},$$

  the radiative-diffusion gradient. Large flux or large opacity gives a steep gradient.
- With the opacity's T, P dependence and hydrostatic equilibrium, $T(\tau)$ becomes a T–P profile.

Gray atmosphere heated from above and below: why two bands?

The profile above is for an atmosphere **heated from below** only. It can represent

- a giant planet whose **interior energy** heats the atmosphere, or
- a rocky planet with a surface, where the incident starlight passes through a **transparent** atmosphere, is absorbed by the ground, and is reradiated upward.

In general, however, **starlight is absorbed throughout the atmosphere**. We keep the same simplifications (gray, LTE, radiative equilibrium, no scattering) and add heating from above (Seager §9.6.3, following Hansen 2008).

The key point: the star is always hotter than the planet. Stellar radiation is therefore absorbed at **visible** wavelengths and reradiated at **infrared** wavelengths, where molecular absorption usually dominates. This suggests two intensities:

- I_{vis}: radiation is only **absorbed**, not emitted;
- I_{ir}: all the absorbed energy is **reradiated**.

The total intensity is $I_{\text{vis}} + I_{\text{ir}}$, and the same addition rule holds for J, F, and K.

Why separate them: radiation is absorbed and emitted at *different altitudes*. Earth's cloudless atmosphere is transparent in the visible (we see the stars at night), so most visible light is absorbed by the ground; at most infrared wavelengths, apart from a few narrow windows, the night sky is opaque.

Heated from above and below: Setup and visible beam

Step 1: radiative equilibrium in two gray bands (Seager §9.6.3; Hansen 2008). With constant opacities $\kappa_{\rm vis}$ and $\kappa_{\rm ir}$, and no visible emission ($S_{\rm vis} = 0$), the equilibrium condition $\int \kappa_\nu (J_\nu - S_\nu)\,\mathrm{d}\nu = 0$ balances visible plus infrared absorption against infrared emission:

$$\kappa_{\rm vis} J_{\rm vis} + \kappa_{\rm ir} J_{\rm ir} = \varepsilon_{\rm ir} = \kappa_{\rm ir} S_{\rm ir} \quad \Longrightarrow \quad \boxed{S_{\rm ir} = J_{\rm ir} + \gamma J_{\rm vis}, \qquad \gamma \equiv \frac{\kappa_{\rm vis}}{\kappa_{\rm ir}}}$$

Write $\tau \equiv \tau_{\rm ir}$ so that $\mathrm{d}\tau_{\rm vis} = \gamma\,\mathrm{d}\tau$.

Step 2: the downward stellar beam. With $S_{\rm vis} = 0$, the visible transfer equation is simply $\mu\,\mathrm{d}I_{\rm vis}/\mathrm{d}\tau_{\rm vis} = I_{\rm vis}$. Collimated starlight enters from direction (θ_0, ϕ_0) with $\mu_0 = \cos\theta_0$ (where the positive z direction is upward):

$$I_{\rm vis}(\tau, \theta, \phi) = I_0\, \mathrm{e}^{-\gamma\tau/\mu_0}\, \delta(\cos\theta + \mu_0)\, \delta(\phi - \phi_0) \quad \big[\text{or } I(\theta,\phi) = I_0\, \delta(\theta - \theta_0)\, \delta(\phi - \phi_0)\big].$$

Its mean intensity and downward flux follow immediately by integrating over $\mathrm{d}\Omega$:

$$J_{\rm vis}(\tau) = \frac{I_0}{4\pi}\, \mathrm{e}^{-\gamma\tau/\mu_0}, \qquad F_{\rm vis}(\tau) = -\mu_0 I_0\, \mathrm{e}^{-\gamma\tau/\mu_0} \quad \text{(downward)}.$$

Heated from above and below: Infrared moments

Step 3: the infrared transfer equation. Substituting $S_{\mathrm{ir}} = J_{\mathrm{ir}} + \gamma J_{\mathrm{vis}}$ into the gray IR transfer equation gives

$$\mu \frac{\mathrm{d}I_{\mathrm{ir}}}{\mathrm{d}\tau} = I_{\mathrm{ir}} - S_{\mathrm{ir}} = I_{\mathrm{ir}} - J_{\mathrm{ir}} - \gamma J_{\mathrm{vis}}.$$

Step 4: zeroth moment ($\times \frac{1}{4\pi} \oint \mathrm{d}\Omega$). Since J_{ir} and J_{vis} are isotropic,

$$\frac{1}{4\pi} \frac{\mathrm{d}F_{\mathrm{ir}}}{\mathrm{d}\tau} = J_{\mathrm{ir}} - S_{\mathrm{ir}} = -\gamma J_{\mathrm{vis}} = -\frac{\gamma I_0}{4\pi} \mathrm{e}^{-\gamma\tau/\mu_0}.$$

Integrating over τ (or directly from total flux conservation $F_{\mathrm{ir}} + F_{\mathrm{vis}} = F_{\mathrm{int}} = \sigma T_{\mathrm{int}}^4 = \mathrm{const}$),

$$F_{\mathrm{ir}}(\tau) = -F_{\mathrm{vis}}(\tau) + F_{\mathrm{int}} = \mu_0 I_0 \mathrm{e}^{-\gamma\tau/\mu_0} + \sigma T_{\mathrm{int}}^4.$$

Step 5: first moment ($\times \frac{1}{4\pi} \oint \mu \, \mathrm{d}\Omega$). The isotropic terms $(J_{\mathrm{ir}} + \gamma J_{\mathrm{vis}}) \oint \mu \, \mathrm{d}\Omega$ vanish:

$$\frac{\mathrm{d}K_{\mathrm{ir}}}{\mathrm{d}\tau} = \frac{1}{4\pi} F_{\mathrm{ir}}(\tau) = \frac{1}{4\pi} \left(\mu_0 I_0 \mathrm{e}^{-\gamma\tau/\mu_0} + \sigma T_{\mathrm{int}}^4 \right).$$

Integrating over τ and applying the Eddington closure $K_{\mathrm{ir}} = \frac{1}{3} J_{\mathrm{ir}}$ gives

$$J_{\mathrm{ir}}(\tau) = 3K_{\mathrm{ir}}(\tau) = \frac{3}{4\pi} \left(\sigma T_{\mathrm{int}}^4 \, \tau - \frac{\mu_0^2}{\gamma} I_0 \mathrm{e}^{-\gamma\tau/\mu_0} \right) + 3c.$$

Heated from above and below: Boundary condition

Step 6: fix the integration constant c from the emergent IR flux. At the top of the atmosphere ($\tau = 0$), no IR radiation enters from space, and the outgoing intensity is $I_{\rm ir}(0,\mu) = \int_0^\infty S_{\rm ir}(\tau)\,\mathrm{e}^{-\tau/\mu}\,\mathrm{d}\tau/\mu$. The upward IR flux at $\tau = 0$ is therefore (Seager Eq. 9.58)

$$F_{\rm ir}(0) = 2\pi \int_0^1 I_{\rm ir}(0,\mu)\,\mu\,\mathrm{d}\mu = 2\pi \int_0^1 \int_0^\infty S_{\rm ir}(\tau)\,\mathrm{e}^{-\tau/\mu}\,\mathrm{d}\tau\,\mathrm{d}\mu.$$

Substitute $S_{\rm ir}(\tau) = J_{\rm ir}(\tau) + \gamma J_{\rm vis}(\tau)$ using Steps 2 and 5:

$$S_{\rm ir}(\tau) = \frac{3\sigma T_{\rm int}^4}{4\pi}\,\tau + 3c + \frac{I_0}{4\pi}\left(\gamma - \frac{3\mu_0^2}{\gamma}\right)\mathrm{e}^{-\gamma\tau/\mu_0}.$$

Carry out the τ-integrals with $\int_0^\infty \tau\,\mathrm{e}^{-\tau/\mu}\mathrm{d}\tau = \mu^2$, $\int_0^\infty \mathrm{e}^{-\tau/\mu}\mathrm{d}\tau = \mu$, and $\int_0^\infty \mathrm{e}^{-\tau(1/\mu+\gamma/\mu_0)}\mathrm{d}\tau = \mu\left(1+\frac{\gamma\mu}{\mu_0}\right)^{-1}$:

$$F_{\rm ir}(0) = \frac{1}{2}\sigma T_{\rm int}^4 + 3\pi c + \frac{I_0}{2}\left(\gamma - \frac{3\mu_0^2}{\gamma}\right)\int_0^1 \frac{\mu^2\,\mathrm{d}\mu}{1+\gamma\mu/\mu_0}.$$

Equating this to the required top flux from Step 4, $F_{\rm ir}(0) = \mu_0 I_0 + \sigma T_{\rm int}^4$, determines $3c$.

Heated from above and below: The temperature profile $T(\tau)$

Step 7: solve for $T(\tau)$ via LTE ($S_{\rm ir} = \sigma T^4/\pi$). With $I_0 \equiv \sigma T_0^4$, the constant $3c$ can be fixed in two slightly different ways:

- **Eddington boundary condition** $J_{\rm ir}(0) = F_{\rm ir}(0)/2\pi$ (at $\tau = 0$, $I_{\rm ir} = 0$ for $\mu < 0$ and $I_{\rm ir} \approx I^+$ is isotropic for $\mu > 0$, so $J_{\rm ir}(0) = \frac{1}{2}\int_0^1 I^+ \mathrm{d}\mu = \frac{1}{2} I^+$ and $F_{\rm ir}(0) = 2\pi \int_0^1 I^+ \mu \,\mathrm{d}\mu = \pi I^+$): this gives $3c = \frac{\sigma}{2\pi}\left[T_{\rm int}^4 + \mu_0 T_0^4 \left(1 + \frac{3\mu_0}{2\gamma}\right)\right]$, and $\frac{\sigma T^4}{\pi} = S_{\rm ir} = J_{\rm ir} + \gamma J_{\rm vis}$ yields

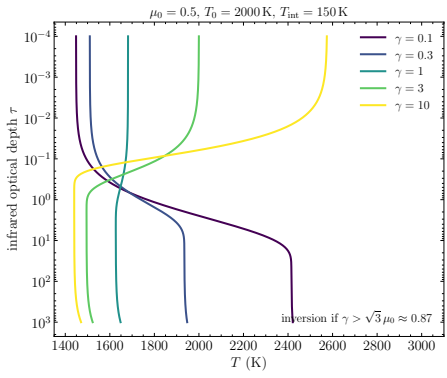
$$T^4(\tau) = \frac{3}{4} T_{\rm int}^4 \left(\tau + \frac{2}{3}\right) + \frac{\mu_0 T_0^4}{4}\left[2 + \frac{3\mu_0}{\gamma} + \left(\frac{\gamma}{\mu_0} - \frac{3\mu_0}{\gamma}\right) \mathrm{e}^{-\gamma\tau/\mu_0}\right]$$

- **Exact emergent-flux integral** (Seager Eqs. 9.58–9.62; Hansen 2008): evaluating $\int_0^1 \frac{\mu^2 \,\mathrm{d}\mu}{1+\gamma\mu/\mu_0}$ explicitly in Step 6 gives

$$T^4(\tau) \approx \frac{3}{4} T_{\rm int}^4 \left(\tau + \frac{2}{3}\right) + \frac{3}{4}\mu_0 T_0^4 \left[\frac{2}{3} + \frac{\mu_0}{\gamma} - \frac{\mu_0}{\gamma} \mathrm{e}^{-\gamma\tau/\mu_0}\right] - \frac{1}{2} T_0^4 \left(\frac{\gamma}{\mu_0} - \frac{3\mu_0}{\gamma}\right) \mu_0^2 \left[\frac{1}{2} - \frac{\mu_0}{\gamma} + \frac{\mu_0^2}{\gamma^2} \ln\left(1 + \frac{\gamma}{\mu_0}\right)\right]$$

Note: Seager's Eq. (9.60) sets $\sigma T^4/\pi = J_{\rm ir}(\tau)$ in the last step, omitting the $+\gamma J_{\rm vis}(\tau)$ term from $S_{\rm ir} = J_{\rm ir} + \gamma J_{\rm vis}$ (Eq. 9.48); keeping it adds the $\frac{\gamma}{\mu_0} \mathrm{e}^{-\gamma\tau/\mu_0}$ term.

Irradiated gray τ-T profiles



The solution with the Eddington boundary condition $(J_{\mathrm{ir}}(0) = F_{\mathrm{ir}}(0)/2\pi)$ for several γ. Script `scripts/make_irradiated_gray_profiles.py`.

- **Deep layers** ($\gamma\tau/\mu_0 \gg 1$): the starlight has been absorbed; $T^4 \to \frac{3}{4} T_{\mathrm{int}}^4 \tau + \mathrm{const}$ — the interior-heated profile, nearly isothermal when T_{int} is small.
- **Visible transparent** ($\gamma \to 0$): the profile tends to $\frac{3}{4}(T_{\mathrm{int}}^4 + \mu_0 T_0^4)(\tau + \frac{2}{3})$, as if all the flux came from below — the ground-heated case.
- **Thermal inversion** (T increases upward, as $\tau \to 0$) occurs when the coefficient of $\mathrm{e}^{-\gamma\tau/\mu_0}$ is positive:

$$\frac{\gamma}{\mu_0} - \frac{3\mu_0}{\gamma} > 0 \quad \Longleftrightarrow \quad \gamma > \sqrt{3}\,\mu_0 \quad (\kappa_{\mathrm{vis}} > \sqrt{3}\,\mu_0\,\kappa_{\mathrm{ir}}).$$

  Physically: strong optical absorbers (TiO/VO, metals, ozone, hazes) trap starlight high in the atmosphere (large κ_{vis}), where the gas cannot cool efficiently in the infrared (small κ_{ir}), forcing the upper layers to heat up.
- Smaller μ_0 (grazing starlight, near the terminator) deposits the energy higher and lowers the deep temperature.

Irradiated gray T–P profiles

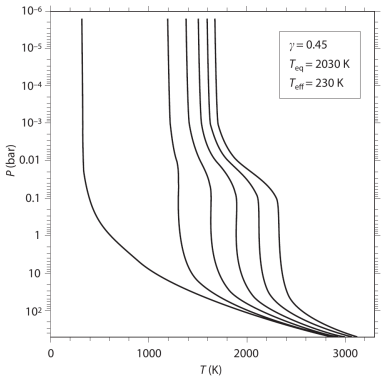


Figure 3: Gray-atmosphere T–P profiles for $\mu_0 = 0.0$–1.0 (left to right), $\gamma = 0.45$. Seager (2010), Fig. 9.7, after Hansen (2008).

- Hydrostatic equilibrium ($\mathrm{d}\tau = \kappa_m \, \mathrm{d}P/g$) converts $T(\tau)$ into a T–P profile.
- **Three altitude regions** (here $\gamma = 0.45 < \sqrt{3}\,\mu_0$, so no inversion):
  1. **Top ($P \lesssim 10^{-3}$ bar, $\tau \ll 1$):** optically thin $\rightarrow$ **isothermal** skin layer.
  2. **Middle (10^{-3}–10 bar):** starlight is absorbed around $\tau_{\rm vis} = \gamma\tau/\mu_0 \sim 1$ (T rises), then settles into a second **isothermal** zone once $\tau_{\rm vis} \gg 1$.
  3. **Deep ($P \gtrsim 10$ bar, $\tau \gg 1$):** interior flux ($T_{\rm int}$) dominates $\rightarrow$ T rises with depth ($T^4 \propto \tau$).
- **Incident angle** $\mu_0 = \cos\theta_0$ (left $\mu_0 = 0$ to right $\mu_0 = 1$): more vertical starlight brings larger flux ($\propto \mu_0$) and penetrates deeper, warming all levels.
- Widely used as the **initial guess** for full radiative-convective models.

The Adiabatic Temperature Profile: Convection basics

Convection transports energy by **bulk vertical motion**: heat flows from hot to cold regions with the moving gas. It is driven by the temperature gradient and enabled by gravity (buoyancy).

- A parcel heated from below becomes lighter than its surroundings and rises. At lower ambient pressure it expands and releases heat; now cooler, it sinks, is reheated, and the cycle repeats.
- Everyday examples: boiling water, lava lamps, the shimmering air above hot pavement.
- The Sun's **granulation**: bright centers are rising columns of hot gas, dark lanes are cooled gas sinking; cells are ~ 1000 km across.

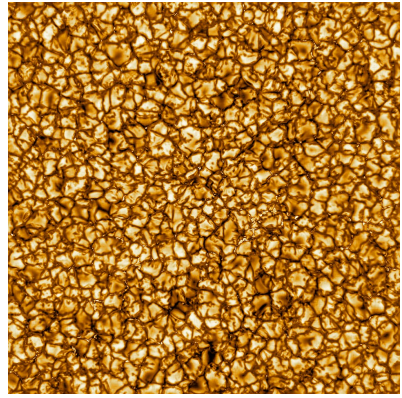


Figure 4: Solar granulation imaged by the Daniel K. Inouye Solar Telescope (NSO/AURA/NSF). Each cell is ~ 1000 km across.

Thermodynamics of an Ideal Gas: Internal Energy

- **First Law of Thermodynamics** (energy conservation):

  Heat added to a system = Change in internal energy + Work done on surroundings

  $$\boxed{\mathrm{d}Q = \mathrm{d}U + P\,\mathrm{d}V}$$

- **Internal energy per particle** of an ideal gas (N = number of particles):

$U/N = \frac{3}{2}\,k_\mathrm{B}T$	for a monatomic gas	(translation about 3 axes)
$U/N = \frac{5}{2}\,k_\mathrm{B}T$	for a diatomic gas	(+ rotation about 2 axes)
$U/N = 3\,k_\mathrm{B}T$	for a polyatomic gas	(+ rotation about 3 axes)

  An ideal gas is a theoretical gas composed of many randomly moving point particles whose only interactions are perfectly elastic collisions (no viscosity or heat conduction).

- **In general**, the internal energy per particle is

  $$\boxed{U/N = \frac{f}{2}\,k_\mathrm{B}T \qquad (f = \text{degrees of freedom})}$$

  At high temperatures, molecules gain access to an increasing number of vibrational degrees of freedom as they start to bend and stretch.

Thermodynamics of an Ideal Gas: Specific Heat Capacities

- **Ideal gas law** (equation of state for a Maxwellian distribution): $\boxed{PV = Nk_\mathrm{B}T}$ or $P = \dfrac{N}{V}k_\mathrm{B}T$.
- **Specific heat capacity** is the heat energy required to raise the temperature of a material per unit mass (M = total mass, $m = M/N = \mu_m m_\mathrm{H}$ = mass per particle, $R_s = k_\mathrm{B}/m$ = specific gas constant):
  - **at constant volume** ($\mathrm{d}V = 0$):

$$c_v \equiv \frac{1}{M}\left(\frac{\partial Q}{\partial T}\right)_V = \frac{1}{M}\left(\frac{\partial U}{\partial T}\right)_V \quad \Longrightarrow \quad \boxed{c_v = \frac{f}{2}\frac{k_\mathrm{B}}{m} = \frac{f}{2}R_s}$$

  - **at constant pressure** ($\mathrm{d}P = 0$):

$$c_p \equiv \frac{1}{M}\left(\frac{\partial Q}{\partial T}\right)_P = \frac{1}{M}\left(\frac{\partial U}{\partial T}\right)_P + \frac{P}{M}\left(\frac{\partial V}{\partial T}\right)_P = \frac{1}{M}\frac{f}{2}Nk_\mathrm{B} + \frac{P}{M}\frac{Nk_\mathrm{B}}{P}$$

$$\Longrightarrow \quad \boxed{c_p = \frac{f+2}{2}\frac{k_\mathrm{B}}{m} = c_v + \frac{k_\mathrm{B}}{m} = c_v + R_s}$$

- **Ratio of specific heat capacities** (**adiabatic index** γ_ad; $c_p > c_v$ since part of the added heat at constant P goes into $P\,\mathrm{d}V$ expansion work):

$$\boxed{\gamma_\mathrm{ad} \equiv \frac{c_p}{c_v} = \frac{f+2}{f}} \quad \Longrightarrow \quad \gamma_\mathrm{ad} = \begin{cases} 5/3 & \text{monatomic } (f=3) \\ 7/5 & \text{diatomic } (f=5) \\ 4/3 & \text{polyatomic } (f=6) \end{cases}$$

The dry adiabatic temperature gradient: First Law of Thermodynamics

Consider an air parcel of mass M, volume V, density $\rho = M/V$, temperature T, and pressure P that expands or compresses **adiabatically** while rising in a hydrostatic atmosphere (Seager §9.7.2).

Step 1: First Law of Thermodynamics (conservation of energy for the parcel):

$$\mathrm{d}U = \mathrm{d}Q - \mathrm{d}W,$$

where $\mathrm{d}U$ is the change in internal energy, $\mathrm{d}Q$ is the heat added to the parcel, and $\mathrm{d}W$ is the work done *by* the parcel on its surroundings.

- **Adiabatic process:** no heat is exchanged with the surroundings, so $\mathrm{d}Q = 0$.
- **Internal energy** of an ideal gas of mass M:

  $$\mathrm{d}U = Mc_v\,\mathrm{d}T = M(c_p - R_s)\,\mathrm{d}T,$$

  where c_v and c_p are the specific heat capacities (per unit mass) at constant volume and constant pressure, and $R_s = k_\mathrm{B}/(\mu_m m_\mathrm{H}) = c_p - c_v$ is the specific gas constant (μ_m: mean molecular weight, m_H: hydrogen mass; c_v, c_p, R_s all in $\mathrm{J\,kg^{-1}K^{-1}}$).
- **Work done by expansion:** $\mathrm{d}W = P\,\mathrm{d}V$. Differentiating the ideal gas law $PV = MR_sT$ (with M constant) gives $P\,\mathrm{d}V + V\,\mathrm{d}P = MR_s\,\mathrm{d}T$, so with $V = M/\rho$,

  $$\mathrm{d}W = P\,\mathrm{d}V = MR_s\,\mathrm{d}T - V\,\mathrm{d}P = MR_s\,\mathrm{d}T - \frac{M}{\rho}\,\mathrm{d}P.$$

The “dry” adiabatic temperature gradient

Step 2: Combine into $\mathrm{d}U = -\mathrm{d}W$ (dividing out the parcel mass M):

$$M(c_p - R_s)\,\mathrm{d}T = -MR_s\,\mathrm{d}T + \frac{M}{\rho}\,\mathrm{d}P \quad \Longrightarrow \quad \boxed{\frac{\mathrm{d}T}{\mathrm{d}P} = \frac{1}{\rho c_p}}$$

Step 3: Convert pressure to altitude z via hydrostatic equilibrium ($\mathrm{d}P = -\rho g\,\mathrm{d}z$, where g is the gravitational acceleration; the parcel always shares the ambient pressure P, while keeping its own T and ρ):

$$\frac{\mathrm{d}T}{\mathrm{d}z} = \frac{\mathrm{d}T}{\mathrm{d}P}\frac{\mathrm{d}P}{\mathrm{d}z} = \frac{1}{\rho c_p}(-\rho g) = -\frac{g}{c_p} \quad \Longrightarrow \quad \boxed{\Gamma_{\mathrm{dry}} \equiv -\frac{\mathrm{d}T}{\mathrm{d}z} = \frac{g}{c_p}}$$

where $\Gamma_{\mathrm{dry}} > 0$ is the **dry adiabatic temperature gradient** (or **adiabatic lapse rate**).

- **In terms of optical depth** ($\mathrm{d}\tau = -\kappa\,\mathrm{d}z$, with extinction $\kappa = \rho\kappa_m$ per unit length and κ_m per unit mass):
  $\Gamma_\tau \equiv \dfrac{\mathrm{d}T}{\mathrm{d}\tau} = \dfrac{g}{\kappa(\tau)\,c_p}$.
- **In terms of pressure** (substituting $\rho = P/(R_s T)$ into $\mathrm{d}T/\mathrm{d}P = 1/(\rho c_p)$):

  $$\frac{\mathrm{d}T}{T} = \frac{R_s}{c_p}\frac{\mathrm{d}P}{P} \quad \Longrightarrow \quad \boxed{T \propto P^{R_s/c_p}} \qquad \left(\frac{R_s}{c_p} = \frac{2}{7} \text{ for a diatomic ideal gas}\right),$$

  so an adiabat is a straight line on a $\log T$–$\log P$ diagram.

The "moist" adiabatic temperature gradient

- **Latent heat:** the thermal energy absorbed or released by a substance during a **phase transition** (e.g. evaporation or condensation) **at constant temperature**. The **specific latent heat** L ($\mathrm{J\,kg^{-1}}$) is the latent heat per unit mass of the phase-changing substance (for water vapor at $0\,^\circ\mathrm{C}$, $L \simeq 2.5 \times 10^6\ \mathrm{J\,kg^{-1}}$).
- When a rising parcel (of dry-gas mass M) cools enough for a condensable vapor (e.g. H_2O on Earth) to condense, the released latent heat warms the parcel:

$$\mathrm{d}Q = -ML\,\mathrm{d}m_s > 0 \qquad (\text{or } \mathrm{d}q \equiv \mathrm{d}Q/M = -L\,\mathrm{d}m_s \text{ per unit mass; } \mathrm{d}m_s < 0 \text{ as vapor condenses}),$$

where $m_s \equiv \frac{\text{mass of condensing vapor}}{\text{mass of dry gas}}$ is the mass mixing ratio of saturating vapor: the mass of condensing vapor per unit mass of dry gas ($\approx$ per unit total mass for a trace condensable vapor).

Derivation: Keeping $\mathrm{d}Q = -ML\,\mathrm{d}m_s$ in the First Law ($\mathrm{d}U = \mathrm{d}Q - \mathrm{d}W$) and dividing by M:

$$(c_p - R_s)\,\mathrm{d}T = -L\,\mathrm{d}m_s - \left(R_s\,\mathrm{d}T - \frac{1}{\rho}\,\mathrm{d}P\right) \quad \Longrightarrow \quad c_p\,\mathrm{d}T - \frac{1}{\rho}\,\mathrm{d}P = -L\,\mathrm{d}m_s.$$

Using hydrostatic equilibrium ($\mathrm{d}P = -\rho g\,\mathrm{d}z$) and the chain rule $\mathrm{d}m_s = (\mathrm{d}m_s/\mathrm{d}T)\,\mathrm{d}T$:

$$\left(c_p + L\frac{\mathrm{d}m_s}{\mathrm{d}T}\right)\mathrm{d}T = -g\,\mathrm{d}z \quad \Longrightarrow \quad \boxed{\Gamma_{\text{moist}} \equiv -\frac{\mathrm{d}T}{\mathrm{d}z} = \frac{g}{c_p}\,\frac{1}{1 + (L/c_p)(\mathrm{d}m_s/\mathrm{d}T)}}$$

The moist adiabatic temperature gradient: Earth and exoplanets

$$\Gamma_{\text{moist}} = \frac{g}{c_p} \frac{1}{1 + (L/c_p)(\mathrm{d}m_s/\mathrm{d}T)} < \Gamma_{\text{dry}} = \frac{g}{c_p}$$

- Because condensation releases latent heat as the parcel rises, a moist parcel cools **more slowly** with altitude than a dry parcel ($\Gamma_{\text{moist}} < \Gamma_{\text{dry}}$).
- **Earth's troposphere:**
  - dry adiabatic temperature gradient: $\Gamma_{\text{dry}} \simeq 9.8\,\text{K}\,\text{km}^{-1}$;
  - moist adiabatic temperature gradient: $\Gamma_{\text{moist}} \simeq 5\,\text{K}\,\text{km}^{-1}$;
  - typical average: $\simeq 6.5\,\text{K}\,\text{km}^{-1}$ (convection is partly moist and partly dry).
- Remarkably, Earth's radiative temperature profile is *stable* against dry convection, but *unstable* against moist convection — **latent heat release is what drives convection in Earth's troposphere**.
- **Exoplanets:** the same formula applies to whatever species condenses at the planet's temperature (e.g. NH_3, CH_4, H_2O, silicates, iron), using its own latent heat L and saturation curve $m_s(T)$.

The One-Dimensional T–P Profile: Conceptual overview

To determine the vertical thermal structure of an atmosphere in **radiative–convective equilibrium** (Seager §9.8.1), we solve **three equations for three unknowns** — $T(\tau)$, $P(\tau)$, and $I(\tau,\mu,\nu)$:

1. **hydrostatic equilibrium:** $\dfrac{\mathrm{d}P}{\mathrm{d}\tau} = \dfrac{g\rho}{\kappa} = \dfrac{g}{\kappa_m}$;
2. **radiative transfer equation:** $\mu\,\dfrac{\mathrm{d}I(\tau,\mu,\nu)}{\mathrm{d}\tau} = I(\tau,\mu,\nu) - S(\tau,\mu,\nu)$;
3. **energy equilibrium** in each layer, comparing the atmosphere's radiative temperature gradient $(-\mathrm{d}T/\mathrm{d}z)_{\mathrm{rad}}$ with a rising parcel's adiabatic gradient $\Gamma_{\mathrm{ad}} \equiv (-\mathrm{d}T/\mathrm{d}z)_{\mathrm{ad}}$ (Γ_{dry} or Γ_{moist}):
   - **radiative equilibrium** $\left[\text{convectively stable, } (-\mathrm{d}T/\mathrm{d}z)_{\mathrm{rad}} < \Gamma_{\mathrm{ad}}\right]$: $\dfrac{\mathrm{d}F_{\mathrm{rad}}}{\mathrm{d}\tau} = 4\pi\int_0^\infty \kappa_\nu (J_\nu - S_\nu)\,\mathrm{d}\nu = 0$;
   - **convective equilibrium** $\left[\text{convectively unstable, } (-\mathrm{d}T/\mathrm{d}z)_{\mathrm{rad}} > \Gamma_{\mathrm{ad}}\right]$: $-\dfrac{\mathrm{d}T}{\mathrm{d}z} = \Gamma_{\mathrm{ad}}$

     $\left(\text{or } \dfrac{\mathrm{d}\ln T}{\mathrm{d}\ln P} = \nabla_{\mathrm{ad}} = \dfrac{R_s}{c_p} \text{ for a dry gas}\right)$.

Iterative solution procedure:

- **Step 1 (Initial guess):** Start from an analytical gray T–P profile (§9.6.3) using T- and P-dependent opacities $\kappa(T,P,\nu)$.
- **Step 2 (Radiative iteration):** Solve the three equations numerically, applying temperature corrections $\Delta T(\tau)$ until a radiative-equilibrium T–P profile satisfies all three equations.
- **Step 3 (Convective adjustment):** Test every altitude for convective stability; where unstable, replace $\mathrm{d}T/\mathrm{d}z$ with the adiabat $-\Gamma_{\mathrm{ad}}$ and iterate again so the radiative zone joins the adiabat smoothly.

A radiative-equilibrium atmosphere: Equations, inputs, and outputs

In the radiative-equilibrium layers (Seager §9.8.2, Eqs. 9.71–9.73), we solve simultaneously:

$$\frac{\mathrm{d}P}{\mathrm{d}\tau} = \frac{g\rho}{\kappa}, \qquad \mu\,\frac{\mathrm{d}I(\tau,\mu,\nu)}{\mathrm{d}\tau} = I(\tau,\mu,\nu) - S(\tau,\mu,\nu), \qquad \frac{\mathrm{d}F(\tau)}{\mathrm{d}\tau} = 4\pi\int_0^\infty \kappa(\tau,\nu)\big[J(\tau,\nu) - S(\tau,\nu)\big]\,\mathrm{d}\nu = 0$$

- **Outputs (3 unknowns solved simultaneously):**
  - Temperature profile $T(\tau)$ and pressure profile $P(\tau)$ $\implies$ the vertical **T–P profile**;
  - Radiation field $I(\tau,\mu,\nu)$ at all depths $\implies$ the **emergent planetary spectrum** $I(0,\mu,\nu)$ at the top of the atmosphere comes out simultaneously with the T–P profile.
- **Inputs (model parameters and boundary conditions):**
  - **Surface gravity** g**:** known directly for transiting exoplanets (from measured mass M_p and radius R_p).
  - **Interior temperature** $T_{\rm int}$**:** sets the upward internal heat flux $F_{\rm int} = \sigma_{\rm R} T_{\rm int}^4$ from the deep interior; taken from planetary thermal evolution models or treated as a free parameter.
  - **Incident stellar irradiation:** top-of-atmosphere downward beam $I_0(\nu)$ along direction $-\mu_0$.
  - **Opacities** $\kappa(\tau,\nu)$**:** depend on the chemical composition; requires input **elemental abundances** together with either **thermochemical equilibrium** or a **photochemical model** (Chapter 4).

A radiative-equilibrium atmosphere: Why iteration is required

Why can't we solve the three radiative-equilibrium equations in a single step (Seager §9.8.2)?

- **Subtlety 1: Tight nonlinear coupling among T, P, chemistry, and I_ν.**
  - Suppose we solve the radiative transfer equation alone for a trial $T(\tau)$, find that radiative equilibrium is violated ($\mathrm{d}F/\mathrm{d}\tau \neq 0$), and update $T(\tau)$ to restore radiative equilibrium.
  - Changing $T(\tau)$ alters the **chemical equilibrium number densities** $n_i(T, P)$ of absorbing molecules and atoms.
  - This changes the **gas pressure** $P(\tau)$ (via the equation of state and hydrostatic equilibrium) as well as the **opacities** $\kappa(\tau, \nu)$ and **emissivities** $\varepsilon(\tau, \mu, \nu)$.
  - Modified $\kappa(\tau, \nu)$ and $\varepsilon(\tau, \mu, \nu)$ in turn alter the **radiation field** $I(\tau, \mu, \nu)$ at *all* depths!

- **Subtlety 2: Non-locality introduced by scattering.**
  - When scattering is present in the source function $S(\tau, \mu, \nu)$, photons are not absorbed and re-emitted locally, so the radiation field $I(\tau, \mu, \nu)$ decouples from the local Planck function $B_\nu(T)$.
  - For an irradiated exoplanet, incident stellar photons can scatter multiple times and travel deep into the atmosphere before being absorbed and heating the gas — directly coupling the top of the atmosphere to the deep layers.

- **Consequence:** A simultaneous, iterative numerical solution is essential, starting from a physically reasonable initial guess (such as the analytical gray T–P profile).

Radiative–convective equilibrium: Convective adjustment

Once a radiative-equilibrium profile $T_{\rm rad}(z)$ is obtained, we check every layer for convective stability using the **Schwarzschild criterion** (Seager §9.8.3, Eq. 9.74):

$$\left|\frac{\mathrm{d}T}{\mathrm{d}z}\right|_{\rm rad} > \left|\frac{\mathrm{d}T}{\mathrm{d}z}\right|_{\rm ad} \equiv \Gamma_{\rm ad} \qquad \Longleftrightarrow \qquad \text{convectively unstable!}$$

- **Dominant-convection approximation:** In planetary atmosphere models, whenever convection sets in, it is usually assumed to be so efficient that it dominates energy transport and forces the temperature gradient to follow the **adiabatic temperature gradient** directly ($\mathrm{d}T/\mathrm{d}\tau = \Gamma_\tau = g/[\kappa(\tau)c_p]$, Eq. 9.69, which already incorporates hydrostatic equilibrium).
- **Fixing the integration constant (boundary condition):**
  - The adiabatic gradient fixes only the *slope* of $T(z)$ or $T(P)$ (giving a family of parallel adiabats, $T \propto P^{R_s/c_p}$).
  - To make the convective profile consistent with the overlying radiative zone, we match T to the **radiative-equilibrium temperature at the top of the convection zone** (the radiative–convective boundary, RCB), assuming convection carries the full energy flux in the convective zone (Eq. 9.75):

$$F_{\rm conv}(\tau) = \sigma_{\rm R} T_{\rm int}^4 + \mu_0 I_0.$$

Radiative–convective equilibrium: Convective flux and mixing-length theory

In transition regions where **both radiation and convection** transport significant fractions of the energy (Seager §9.8.3), radiative equilibrium is replaced by total flux conservation (Eq. 9.76):

$$\boxed{F_{\rm rad}(\tau) + F_{\rm conv}(\tau) = \sigma_{\rm R} T_{\rm eff}^4 + \mu_0 I_0}$$

(Note: In Seager Eqs. 9.75–9.76, $\sigma_{\rm R} T_{\rm int}^4 + \mu_0 I_0$ or $\sigma_{\rm R} T_{\rm eff}^4 + \mu_0 I_0$ denotes the total outgoing thermal flux when the incident beam $\mu_0 I_0$ is deposited deep down; the net upward minus downward bolometric flux is $\sigma_{\rm R} T_{\rm int}^4$.)

How do we express the convective flux $F_{\rm conv}$? (Eqs. 9.77–9.78)

- A rising convective mass element of density ρ and specific heat c_p that is warmer than its surroundings by ΔT deposits excess thermal energy $\rho c_p \Delta T$ per unit volume when it merges with the ambient atmosphere. Moving upward at mean speed $\bar{v}$, it carries a heat flux

$$F_{\rm conv} = \bar{v} \rho c_p \Delta T, \qquad \Delta T = \left[\left(-\frac{{\rm d}T}{{\rm d}z} \right) - \left(-\frac{{\rm d}T}{{\rm d}z} \right)_{\rm c} \right] \Delta z,$$

  where $(-{\rm d}T/{\rm d}z)$ is the temperature gradient of the surrounding atmosphere (in radiative–convective equilibrium) and $(-{\rm d}T/{\rm d}z)_{\rm c}$ is the internal temperature gradient of the rising convective element over distance Δz.

- **Mixing-length theory (MLT):** While c_p is known and ρ comes from the ideal gas law and hydrostatic equilibrium, finding $\bar{v}$ and Δz requires **mixing-length theory** — a local 1D approximation from stellar astrophysics. MLT assumes a convective bubble rises a characteristic **mixing length** $l = \Delta z \sim H$ (roughly one pressure scale height H) before releasing its heat and sinking.

Is the deep atmosphere convective? Milne's solution and CIA

Write the Schwarzschild criterion in logarithmic form: the atmosphere is convectively stable if

$$\frac{\partial \ln T}{\partial \ln P} \leq \nabla_{\rm ad} = \frac{R_s}{c_p} = \frac{2}{2+f},$$

where f is the number of degrees of freedom of the dominant gas: for a diatomic gas (H_2 on giant planets; N_2 on Earth and Titan), $\nabla_{\rm ad} = 2/7 \approx 0.286$ (translation + rotation, $f = 5$) or $1/4 = 0.25$ (with vibration activated, $f = 6$).

Deep in the atmosphere ($\tau \gg 1$), internal heat gives Milne's radiative solution, $T^4 \propto \tau$:

- **Constant opacity ($\kappa_m = $ const):** $\tau \propto P \implies T \propto P^{1/4}$, so $\dfrac{\partial \ln T}{\partial \ln P} \to \dfrac{1}{4} \leq \dfrac{2}{7}$: **convectively stable** (remains radiative).
- **Collision-induced absorption (CIA) dominates ($\kappa_m \propto P$):** $\tau \propto P^2 \implies T \propto P^{1/2}$, so $\dfrac{\partial \ln T}{\partial \ln P} \to \dfrac{1}{2} > \dfrac{2}{7}$: **unconditionally convective!**

This result is independent of absorption strength, scattering, or gravity g: whenever CIA dominates deep down, the interior of a gas giant *must* convect.

Higher up, strong stellar irradiation tends to stabilize the T–P profile, though a strong greenhouse ($\kappa_{\rm vis} \ll \kappa_{\rm ir}$) can produce detached convective zones between radiative layers.

Temperature Retrieval: Forward vs. inverse problems

- **The forward problem (§9.3–§9.8):**

$$\text{Inputs } \left(g, F_*, \text{abundances}, \kappa_\nu\right) \xrightarrow{\text{iterative solver}} \text{T–P profile } T(P) \longrightarrow \text{Emergent spectrum } F_\nu(0).$$

- **The inverse problem — "Temperature & Abundance Retrieval" (§9.9):**

$$\text{Observed spectrum } F_\nu(0) \xrightarrow{\text{retrieval}} \text{T–P profile } T(P) \text{ and molecular abundances.}$$

- **Solar System planets vs. Exoplanets:**
  - **Solar System planets:** High-resolution, high-S/N spectra allow one to infer a good initial (*fiducial*) T–P profile directly from the data, and then apply small perturbations to $T(P)$ and molecular abundances until the spectrum is fit.
  - **Exoplanets:** Spectra have much lower S/N and spectral coverage, so no unique fiducial T–P profile can be read off from the data. Instead, we must computationally explore the full parameter space to map out the **allowed ranges** of T–P profiles and atomic/molecular abundances.

- **Why a parametric T–P profile is needed (Madhusudhan & Seager 2009):**
  - Self-consistent forward radiative–convective models are far too computationally expensive to run 10^5–10^7 times.
  - Instead, one uses a fast **parametric T–P profile** (run tens of thousands of times or more) while enforcing **hydrostatic equilibrium** and **global energy balance** — constraining the allowed T–P range, molecular abundances, Bond albedo, day–night energy redistribution, and effective temperature $T_{\rm eff}$.

A parametric T–P profile: Mathematical formulation

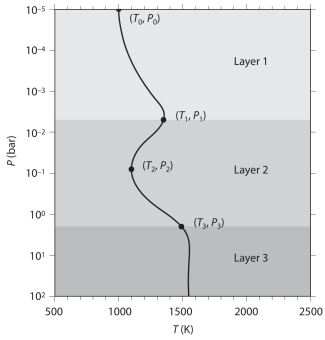


Figure 5: Parametric P–T profile with a thermal inversion in Layer 2 and an isothermal deep Layer 3. Seager (2010), Fig. 9.8, after Madhusudhan & Seager (2009).

Motivated by basic atmospheric physics, Solar System profiles, and 1D forward models, Madhusudhan & Seager (2009) parameterize pressure P (in bar, used as the vertical coordinate) as a generalized exponential of T (in K) across three layers (Seager §9.9, Eqs. 9.79–9.80):

$$\begin{aligned} &\text{Layer 1} \quad (P_0 < P < P_1): \quad P = P_0\, e^{\alpha_1 (T-T_0)^{\beta_1}} \\ &\text{Layer 2} \quad (P_1 < P < P_3): \quad P = P_2\, e^{\alpha_2 (T-T_2)^{\beta_2}} \\ &\text{Layer 3} \quad (P > P_3): \quad T = T_3 \quad \text{(isothermal)} \end{aligned}$$

Counting the free parameters (from 11 down to 6):

- Initially there are 11 parameters: $P_0, T_0, \alpha_1, \beta_1, P_1, P_2, T_2, \alpha_2, \beta_2, P_3, T_3$.
- Empirically β_1 and β_2 are redundant and fixed to $\beta_1 = \beta_2 = 0.5$ (leaving 9 parameters).
- The top of the model atmosphere is fixed at $P_0 = 10^{-5}$ bar (leaving 8 parameters).
- Requiring **temperature continuity** at the two layer boundaries (P_1 between Layers 1 & 2, and P_3 between Layers 2 & 3) eliminates 2 more parameters:

$$T_1 = T_0 + \left[\ln(P_1/P_0)/\alpha_1\right]^2 = T_2 + \left[\ln(P_1/P_2)/\alpha_2\right]^2,$$
$$T_3 = T_2 + \left[\ln(P_3/P_2)/\alpha_2\right]^2.$$

- This leaves **6 free parameters** (e.g. $T_0, \alpha_1, \alpha_2, P_1, P_2, P_3$).

A parametric T–P profile: Physical meaning of each layer

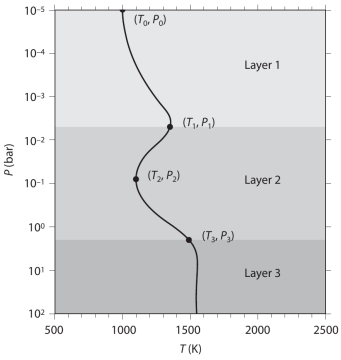


Figure 6: Three-layer parametric P–T profile (Seager 2010, Fig. 9.8).

Each layer in Figure 9.8 reflects the dominant physical regime at that depth (Seager §9.9):

- **Above Layer 1 ($P < P_0 = 10^{-5}$ bar):** Optical depth is very small at all wavelengths ($\tau_\nu \ll 1$); the atmosphere is virtually transparent to incoming and outgoing radiation and produces no significant spectral features.
- **Layer 1 ($P_0 < P < P_1$, upper atmosphere):** No thermal inversion; heated from below by the lower layers and cools upward with increasing altitude (T decreases as P decreases).
- **Layer 2 ($P_1 < P < P_3$, photosphere / stratosphere):** Optically thin-to-moderate region where **most spectral absorption and emission features are formed**. Governed by radiative processes (and atmospheric dynamics). A **thermal inversion** ($P_2 > P_1$, as drawn) can form here when strong stellar irradiation meets strong optical/UV absorbers (gases or haze particles).
- **Layer 3 ($P > P_3$, deep radiative-diffusion zone):** High optical depth ($\tau \gg 1$) prevents stellar radiation from penetrating directly; radiative diffusion yields a nearly **isothermal** structure ($T = T_3$) heated by interior energy outflow. **Below Layer 3**, high pressure/density produces high opacity, making **convection** the dominant transport mechanism.
- **Cooler planets (vs. hot Jupiters):** The isothermal Layer 3 is absent, Layer 2 extends deeper, and the radiative–convective boundary lies at higher altitude (lower P), so convection makes the profile in Layers 2–3 follow an **adiabat**.

Summary: Vertical Thermal Structure (Seager §9.10)

- **Hydrostatic equilibrium & scale height (§9.3):** Pressure supports the atmosphere against gravity, $\mathrm{d}P/\mathrm{d}z = -\rho g$ (or $\mathrm{d}P/\mathrm{d}\tau = g\rho/\kappa$). For an isothermal ideal gas, $P(z) = P_0\,\mathrm{e}^{-z/H}$ with scale height $H = k_\mathrm{B}T/(\mu_m m_\mathrm{H} g)$.
- **Toy greenhouse models (§9.5):** Energy balance gives $T_s = [2/(2-\alpha)]^{1/4} T_e$ for a single leaky infrared-absorbing layer (absorptivity α) and $T_s = (n+1)^{1/4} T_e$ for n opaque layers.
- **Radiative equilibrium & gray T–P profiles (§9.4, §9.6):** When radiation dominates, energy conservation requires $\mathrm{d}F_\mathrm{rad}/\mathrm{d}\tau = 4\pi \int \kappa_\nu (J_\nu - S_\nu)\,\mathrm{d}\nu = 0$.
  - Heated from below only (gray, Eddington): $T^4(\tau) = \frac{3}{4} T_\mathrm{int}^4 (\tau + \frac{2}{3})$.
  - Irradiated two-band gray model ($\gamma = \kappa_\mathrm{vis}/\kappa_\mathrm{ir}$): yields greenhouse warming when $\gamma < \sqrt{3}\mu_0$ and a **thermal inversion** when $\gamma > \sqrt{3}\mu_0$.
- **Convection & adiabatic temperature gradient (§9.4, §9.7):** When radiative transport is inhibited by high opacity or large flux so that $(-\mathrm{d}T/\mathrm{d}z)_\mathrm{rad} > \Gamma_\mathrm{ad}$, **convection** takes over and sets $-\mathrm{d}T/\mathrm{d}z = \Gamma_\mathrm{ad}$:

$$\Gamma_\mathrm{dry} = \frac{g}{c_p} \quad \left(T \propto P^{R_s/c_p}\right), \qquad \Gamma_\mathrm{moist} = \frac{g}{c_p}\,\frac{1}{1 + (L/c_p)(\mathrm{d}m_s/\mathrm{d}T)} < \Gamma_\mathrm{dry}.$$

- **1D radiative–convective models & retrieval (§9.8–§9.9):** Self-consistent forward models iteratively solve hydrostatic equilibrium, radiative transfer, and radiative–convective equilibrium simultaneously for $T(\tau)$, $P(\tau)$, and $I(\tau, \mu, \nu)$. Exoplanet spectral retrievals replace the slow forward solver with a 6-parameter empirical T–P profile to map out allowed temperature structures and molecular abundances.

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